Reliability Engineering and System Safety 244 (2024) 109917

Contents lists available at ScienceDirect

Reliability Engineering and System Safety

journal homepage: www.elsevier.com/locate/ress

A building information modeling-based fire emergency evacuation simulation system for large infrastructures

Zhikun Ding [a,b,c,d], Shengqu Xu [c,e], Xiaofeng Xie [f], Kairui Zheng [g], Daochu Wang [f], Jianhao Fan [f], Hong Li [c], Longhui Liao [a,b,c,d,*]

[a] Key Laboratory for Resilient Infrastructures of Coastal Cities (Shenzhen University), Ministry of Education, China
[b] Guangdong Laboratory of Artificial Intelligence and Digital Economy (SZ), Shenzhen University, Shenzhen, China
[c] Sino-Australia Joint Research Center in BIM and Smart Construction, Shenzhen University, Shenzhen, China
[d] Shenzhen Key Laboratory of Green, Efficient and Intelligent Construction of Underground Metro Station, Shenzhen University, Shenzhen, China
[e] Buji Street Office of Longgang District, Shenzhen, China
[f] Guangzhou Construction Engineering Co., Ltd., Guangzhou, China
[g] Tongyan Digital Intelligence Technology (Chongqing) Co., Ltd., Chongqing, China

ARTICLE INFO

Keywords:
Large infrastructure
Building information modeling
Fire emergency
Evacuation simulation
Optimization
Subway station

ABSTRACT

During fire emergencies, people in large infrastructures are difficult to comprehend real-time fire dynamics and identify available evacuation routes, leading to lengthy evacuation time and unnecessary loss of life and property. This study aims to develop a fire emergency evacuation simulation system for large infrastructures. This system integrates the data from building information modeling platform, fire dynamics simulator, and evacuation simulation platform through customized developments. Real-time fire situation is transmitted to the evacuation simulation platform to assess the impact of dynamic fire spread on the evacuation of people. The behavioral and psychological characteristics of evacuees of different genders and age groups are considered in simulated evacuations. An algorithm for optimizing evacuation route planning is designed to improve the utilization of each evacuation exit and provide a visualization of evacuation routes. This proposed system was validated at a busy subway station in China. The evacuation efficiency increased by 12.87%. It is concluded that this system can provide real-time dynamic optimal evacuation route planning for enhanced emergency management.

1. Introduction

With the emergence of an increasing number of large-scale infrastructures and high-rise buildings, the challenge of emergency rescue from accidents has escalated. As a result, mass deaths and injuries have often occurred [1–3]. Fire is one of the most common and significant disasters that directly threaten human safety. For example, in 2018, domestic fires in the United States (U.S.) caused 3655 deaths, 15,200 injuries, and a direct economic loss of 26.5 billion U.S. dollars [4]. In China, many infrastructures, such as underground subways, are currently under construction, creating potential risks of fire emergencies. Serious subway accidents have occurred in cities such as Beijing and Hong Kong [5,6].

While the emergence of large infrastructures has effectively improved the utilization of urban space [7], their spatial complexity has increased the complexity of effective escape during emergencies [8]. The evacuation in large infrastructures is much more complex than that of other civil buildings and involves a higher pedestrian flow. During a disaster emergency state, it is difficult for people to comprehend the layout of infrastructure and evacuation channels. In the event of a fire disaster, evacuees would experience fear and anxiety due to the sudden changes in their surroundings, including heavy smoke, blaring sirens, high temperatures, and dense crowds. These stimuli can affect the evacuees' rational decision-making during an evacuation and eventually leading to serious loss of life and property [9,10].

Evacuation simulation is an effective method for reproducing the evacuation behaviors of people during fire accidents. It is of great significance to improve the understanding of personnel decision-making during the evacuation process [11]. Users can develop effective strategies to enhance evacuation efficiency and reduce the risks faced by

* Corresponding author at: Key Laboratory for Resilient Infrastructures of Coastal Cities (Shenzhen University), Ministry of Education, China.
E-mail address: liao.longhui@szu.edu.cn (L. Liao).

https://doi.org/10.1016/j.ress.2023.109917
Received 28 January 2023; Received in revised form 8 December 2023; Accepted 26 December 2023
Available online 29 December 2023
0951-8320/

individuals [12]. Building a realistic and efficient evacuation simulation model is challenging due to the uncertainty of fire conditions and individual behaviors during an evacuation [13,14]. For example, smoke and carbon dioxide generated by changing fire situations, physical characteristics of evacuees, and their evacuation speed significantly impact the evacuation process. However, these factors were not fully considered in previous studies. Regarding route selection, priority is given to the shortest escape route. This is likely to cause overcrowding, and there are more efficient routes than the shortest one. Deep learning related intelligent diagnosis and algorithms can be designed to improve computational efficiency and achieve a faster evacuation response [15]. Thus, providing evacuees with a humane, safe, and efficient evacuation route is vital, which needs more investigation.

This study aims to develop a fire emergency evacuation simulation system that can provide real-time, accurate, and efficient evacuation route planning for large infrastructures. The specific purposes are to: (1) establish a fire emergency evacuation model that conforms to objective reality (characteristics of fire dynamics and human behaviors); (2) realize fusion of BIM data, sensor data, and simulation platforms; (3) design an evacuation route optimization and visualization algorithm considering overall utilization of evacuation exits; and (4) enable data integration and interaction between the system and BIM platform. BIM data are introduced into the system to ensure the accuracy of environmental data in infrastructure. Fire Dynamics Simulator (FDS) was used to analyze changes in the fire. The optimization algorithm can adjust fire emergency evacuation routes in real time based on fire dynamics. This ensures that the fire emergency evacuation system can provide real-time and accurate route planning. The proposed system not only ensures the efficient service of evacuation channels, but also reduces secondary damage caused by evacuees' incorrect selection of evacuation routes. It enables operations and maintenance personnel to quickly make evacuation decisions and provide appropriate guidance. Additionally, this system inspires both academics and practitioners to conduct further research and development, contributing to the global body of knowledge on fire emergency evacuation.

The rest of this article is organized as follows. Section 2 reviews research on fire emergency evacuation with emphasis on the use of BIM and simulation technology. Section 3 presents the research methodology. Section 4 describes the design and implementation of the fire emergency evacuation simulation system. Section 5 provides a case study for verifying the applicability of this system. Section 6 presents the results and validation. Section 7 draws a conclusion of this study.

2. Literature review

2.1. Fire emergency evacuation in infrastructures

Fires can be fatal to human life, the environment, and infrastructures, especially in areas with a high population density, such as subway and railway stations. Safe emergency evacuation is an effective way in the event of a fire. Fire emergency evacuation in large infrastructures mainly refers to the process of guiding people affected by a fire inside or around infrastructures to evacuate to a safe area that is not threatened by the fire [16]. Japanese scholars conducted early studies on evacuation indicator lights for fire emergency evacuation systems. For example, by combining the evacuation indicator lighting system with emergency evacuation, effective evacuation guidance was achieved [17]. Moreover, Japanese scholars have integrated early warning information about extreme climate disasters into an emergency evacuation system and disseminated it through social media to ensure that users understand and can make informed decisions during disasters, thereby improving the efficiency of disaster response [18]. Under the influence of the 9/11 terrorist attack and other extreme disasters such as tornadoes, the U.S. has further strengthened research on emergency evaluation systems. Kinateder et al. [19] investigated the impact of exit familiarity and other pedestrians' egress behaviors on evacuees' exit choice in an ambulatory virtual environment, which was critical to pedestrian evacuation modeling and decision-making in the evacuation system. In Germany, previous studies have combined indoor navigation with emergency evacuation systems and proposed personalized indoor navigation evacuation strategies [20]. In China, Liu et al. [21] used a simulated-based multi-objective (fire spread speed, evacuation time, and evacuation path reliability) optimization approach to develop a fire response system model to derive an optimal evacuation strategy. Other studies have also reported that indicator lighting systems with fixed evacuation directions are commonly utilized during fire emergency evacuation. However, these systems are unable to adapt evacuation guidance based on the evolving fire situation, leaving evacuees to estimate safe areas and potentially miss the opportunity for a safe evacuation. When a fire occurs, ensuring the healthy operation of systems is crucial for personal safety. Eliminating system faults through the use of convolutional neural networks is also an important research direction for evacuation systems [22,23].

2.2. Application of BIM in safe evacuation

The application of BIM in the field of safe evacuation is increasing. Previous studies have primarily utilized information, such as materials, geometries, and spacing, contained in three-dimensional (3D) models as a data source for environmental modeling. For example, Tang et al. [24] used the BIM platform as an evacuation management tool, and visualized safety pre-control measures in the form of a database through a self-developed plug-in in the BIM platform, so that it is convenient to safety designers to apply the BIM platform for safety design. Bina and Moghadas [25] combined BIM with agent-based simulation, set and adjusted the characteristics of evacuees, and finally estimated the impact of gender isolation and other characteristics in the evacuation process on safe evacuation in the simulation results. This provides a reference for safety design. Jeong [26] used BIM to create a city information model, which was then integrated with 3D terrain information such as mountains and rivers, and analyzed evacuation plans through algorithms to determine safe routes to refuge during flood disasters. Evacuation routes and visual aids were provided to assist evacuees. Liu [27] used geographic information system and spatial analysis techniques to develop an evacuation accessibility model that integrates and analyzes spatial data such as building and person density to assess the accessibility of different areas and help decision-makers improve evacuation strategies. Therefore, previous studies have verified that the application of BIM tools can help realize the visualization of safe evacuation and data integration between data sources and simulation platforms.

2.3. Application of simulation technology in fire emergency evacuation

An effective method to reproduce evacuees' evacuation behaviors in fire accidents through simulation technology is beneficial for enhancing the understanding of fire emergency management. In recent decades, more and more literature has focused on developing models that can simulate the evacuation of individuals during fire emergencies. According to simulation technology, these models can be divided into three categories: the social force model, cellular automata models, and agent-based models. The social force model was proposed by Helbing and Molnar [28]. This model regards individuals as particles, considering their willingness, mutual relationship, and mutual influence between individuals and the environment as the force for simulation, but the simulation of evacuation in a fire ignores the impact of behavioral decision-making of people [29]. Furthermore, Blue et al. [30] proposed a pedestrian model based on cellular automata, which can change the states of cells according to the rules of human behavior. Liu et al. [31] used cellular automata models to simulate passenger behavior during a ship fire evacuation to analyze the effect of different evacuee densities on evacuation time and routes. Agent-based models can define the characteristics of different evacuees, and naturally describe complex interactions and discrete structures among evacuees. Such models are very flexible and efficient in the definition of characteristics in evacuation simulation [32]. Based on the 2003 Daegu subway fire case in South Korea, Tsukahara et al. [33] used the FDS platform to study effective large-scale evacuation routes during the subway fire. It revealed that evacuation stairs had high temperatures and smoke density, and smoke flow also overlapped with evacuation routes, while downward evacuation may be more effective than upward evacuation for a large-scale subway fire.

2.4. Evacuation route planning

Evacuation route planning is essential to an effective emergency evacuation system. The result of evacuation route planning directly affects the safe evacuation of evacuees. Relevant previous studies mainly focused on extending graph theory and considering superposition analysis of distances among nodes in complex networks to optimize evacuation routes [34–36]. Common route-finding algorithms include but are not limited to heuristic algorithm, A-star algorithm, and Dijkstra algorithm. In addition, some scholars have also conducted route analysis to select multi-node routes. For example, Wang et al. [37] divided a spatial network into hexagonal cells, applied the cellular ant algorithm for passenger ship route planning, and combined heuristic functions to optimize the route planning. To minimize evacuation time and maximize evacuation efficiency, Khakzad [38] used Dijkstra's algorithm and mathematical planning methods to calculate the optimal evacuation route. Liu et al. [21] adopted the Skyline operator to determine the optimal solution for both response time and firefighter allocation using the dataset from the fire emergency response process simulation based on Petri nets at a metro station in Wuhan, China.

Based on research on evacuation route planning algorithms, previous studies [39–41] have combined these algorithms with evacuation simulations to accurately simulate crowd evacuation. This ensures that the results of evacuation route planning are meaningful. The combination enables scholars to comprehensively consider the characteristics of evacuees, such as gender differences and sense of fear.

However, previous studies tended to focus on the selection and coordination of different evacuation passages in buildings, which may not accurately reflect actual fire emergencies, limiting their practicality and usefulness. Therefore, the future research trend is towards the development of an intelligent and personalized fire emergency evacuation simulation system that can track evacuees in real time [42].

2.5. Summary

The literature review revealed the following deficiencies: (1) many previous studies focused on fire emergency evacuation for buildings other than infrastructures which are more unique; (2) most of previous studies on fire emergency evacuation simulation systems adopted preset emergency plans, and ignored the influence of evacuees' characteristics and changes in infrastructures and fire spread on evacuation path planning during fire emergencies; (3) most of existing studies focused on evacuation path planning of small-scale fire scenarios, with evacuation scenarios being simplified, which was not consistent with actual circumstances and may not provide sufficient reference value for practice; and (4) although previous studies have laid the foundation for BIM-based safe evacuation in fire emergencies, few have integrated BIM data, sensor data, and simulation technology for fire emergency evacuation route planning and optimization.

To fill these gaps, the present study has developed an accurate, real-time, intelligent, and personalized fire emergency evacuation simulation system for large infrastructures based on the integration of BIM and sensor data. Real-time indoor temperature, smoke, structural elements, and human behaviors are considered. An optimization algorithm is designed to balance each evacuation exit. Thus, this study contributes to the global body of knowledge on fire emergency evacuation guidance.

3. Research methodology

Fig. 1 shows the research methodology, which was divided into four stages. Firstly, literature analysis was carried out to determine main influencing factors and relevant parameters of fire emergency evacuation, which were then verified through a questionnaire survey. Secondly, a building information model was created to obtain spatial geometric information and key location details such as safe passages and sensor placements. When a fire occurs, evacuees are often prone to panic, which affects their decision-making and judgment during the evacuation process. Evacuees of different ages and genders react differently to fire disaster, leading to distinct behavioral choices regarding evacuation speed, route selection, and other aspects. These were considered and incorporated into the settings of the evacuation simulation platform based on the social force model and evacuation theory. An efficient evacuation strategy can significantly minimize the harm caused by fires. However, the shortest evacuation route in the event of a fire may not always be the most optimal one. The problem of congestion would increase the overall evacuation time. Based on the information model and evacuation simulation dynamics, this study optimized evacuation paths by designing an algorithm that achieves a balance between congestion and evacuation efficiency at each exit. Meanwhile, Java was used for further customized developments, which helps obtain the optimal evacuation path and visualize it in the information model. Thirdly, owing to these efforts, a fire emergency evacuation simulation system was developed to integrate these applications. Finally, a case study was conducted on a large-scale underground subway station to verify the functions of real-time fire monitoring, evacuation route optimization, and output visualization provided by the system to achieve efficient fire emergency evacuation.

4. Design and implementation of the BIM-based fire emergency evacuation simulation system

Guided by BIM, simulation technology, and evacuation theory, a framework for the fire emergency evacuation simulation system is proposed, as shown in Fig. 2. The system is divided into various subsystems for fire monitoring, evacuation visualization, data visualization, and path programming. Simulation modeling of infrastructures and personnel was completed on the simulation platforms based on these functional requirements. Finally, the system is established using both BIM data and sensor data.

The architecture of the system can be divided into three layers: data, implementation, and application. The data layer consists of three parts: BIM database, fire database, and evacuation database. These databases contain the basic data required for the operation of the fire emergency evacuation simulation system. The implementation layer integrates the BIM platform, fire simulation platform, and evacuation simulation platform through the data provided by the data layer to establish an objective and real-time simulation model. Through customized development, these platforms can interact with each other, enabling data exchange and visualization of results. The application layer outputs the results of system operations. It can display fire dynamics, generate evacuation route plans, and provide available evacuation routes and exits. This avoids the risk of not finding an effective evacuation route.

4.1. Integration of BIM and fire simulation

Fire simulation requires extracting geometric properties (such as size and distance) and material information of building components from the information model. This ensures that fire simulation data are combined with structure layout and building materials. This study used PyroSim as the fire simulation platform. PyroSim integrates the functionalities of FDS and Smokeview. It facilitates access to BIM data in IFC, FDS, and

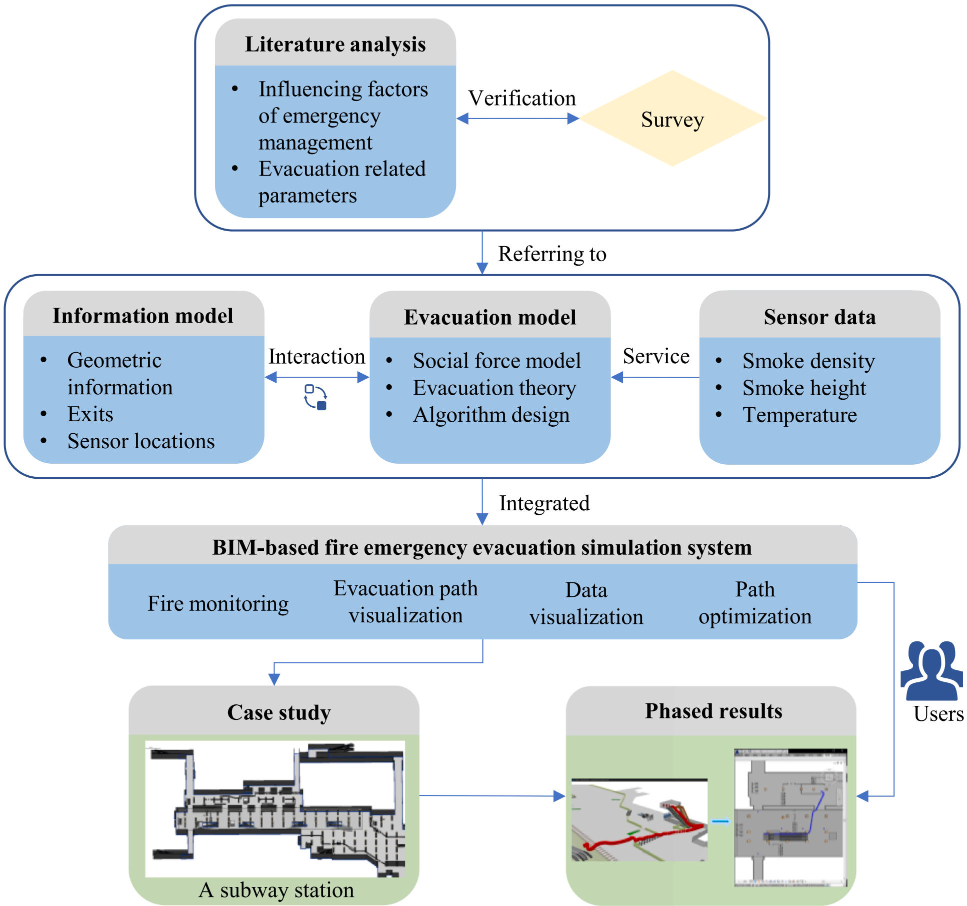


Fig. 1. Research methodology.

other formats, avoiding repeated modeling operations [43]. The model imported into PyroSim carries information such as the original material, making it convenient for users to set material properties. In addition, sensors in the fire simulation are positioned according to their locations in the information model, which ensures data accuracy of fire spread in the subsequent evacuation simulation.

The Revit software provides a variety of data interaction formats on building components' geometric dimensions and material information, such as exporting the *.fbx* format or the *.ifc* format to meet the data format requirements of the PyroSim software for fire spread simulation. However, Revit does not provide coordinates among component properties. To obtain accurate data of temperature and smoke density in the real environment, it is necessary to obtain accurate coordinates of sensors. Fig. 3 shows the use of Dynamo tools [44] to extract internal data of the information model. The coordinates of sensor devices are written into the "Mark" entry of component properties. The sensor information table is summarized to obtain basic sensor attribute information which is then imported into PyroSim.

4.2. Integration of fire simulation and evacuation simulation

This study integrates real-time fire data obtained from the fire simulation platform (PyroSim) to simulate the evacuation of people in infrastructure and improve the authenticity and accuracy of the evacuation process. This study chooses to conduct an evacuation simulation by combining the social force model and agent-based model. AnyLogic was selected as the evacuation simulation platform because it is an open-source software that can simultaneously support discrete and continuous models, system dynamics, and agent-based models [45]. In addition, the AnyLogic software also supports users in utilizing the Java programming language for customized development according to their specific needs, meeting special requirements related to data input and output.

This study uses Java to customize development for fire emergency evacuation on the AnyLogic platform. The integration of fire simulation data and evacuation simulation data has been achieved. When the time value of the evacuation simulation is consistent with the time value of the fire simulation, sensor data from the fire simulation are assigned to sensor agents at corresponding positions in the evacuation model. When sensor data values reach dangerous alarm threshold, marking function of "Polygonal Node" (as shown in Fig. 4) in the evacuation model is used to represent the occurrence and spread of fire, and indicates the impact of the fire's location on pedestrian speed. For example, if smoke sensors detect that smoke height (distance between floor and smoke) is less than 1.5 m, it is determined that evacuation speed in the corresponding area (Smoke spreading area in Fig. 4) would decrease to 90% of normal speed. If the smoke height is lower than 1.2 m, the evacuation speed will decrease to 60%. Generally, when smoke height is less than 1.0 m, evacuees are often unable to move normally, which is regarded as the limit of human body, and the corresponding area (Inaccessible area in Fig. 4) is set to be inaccessible area. For ease of understanding, this study summarizes the implementation description of the algorithm, as shown in Table 1.

4.3. Integration of evacuation simulation and BIM

Evacuation simulation also requires an information model as a reference for the simulation model to ensure the accuracy of visualized scenes and consistency between the simulation and the actual environment. This helps ensure that the evacuation simulation accurately

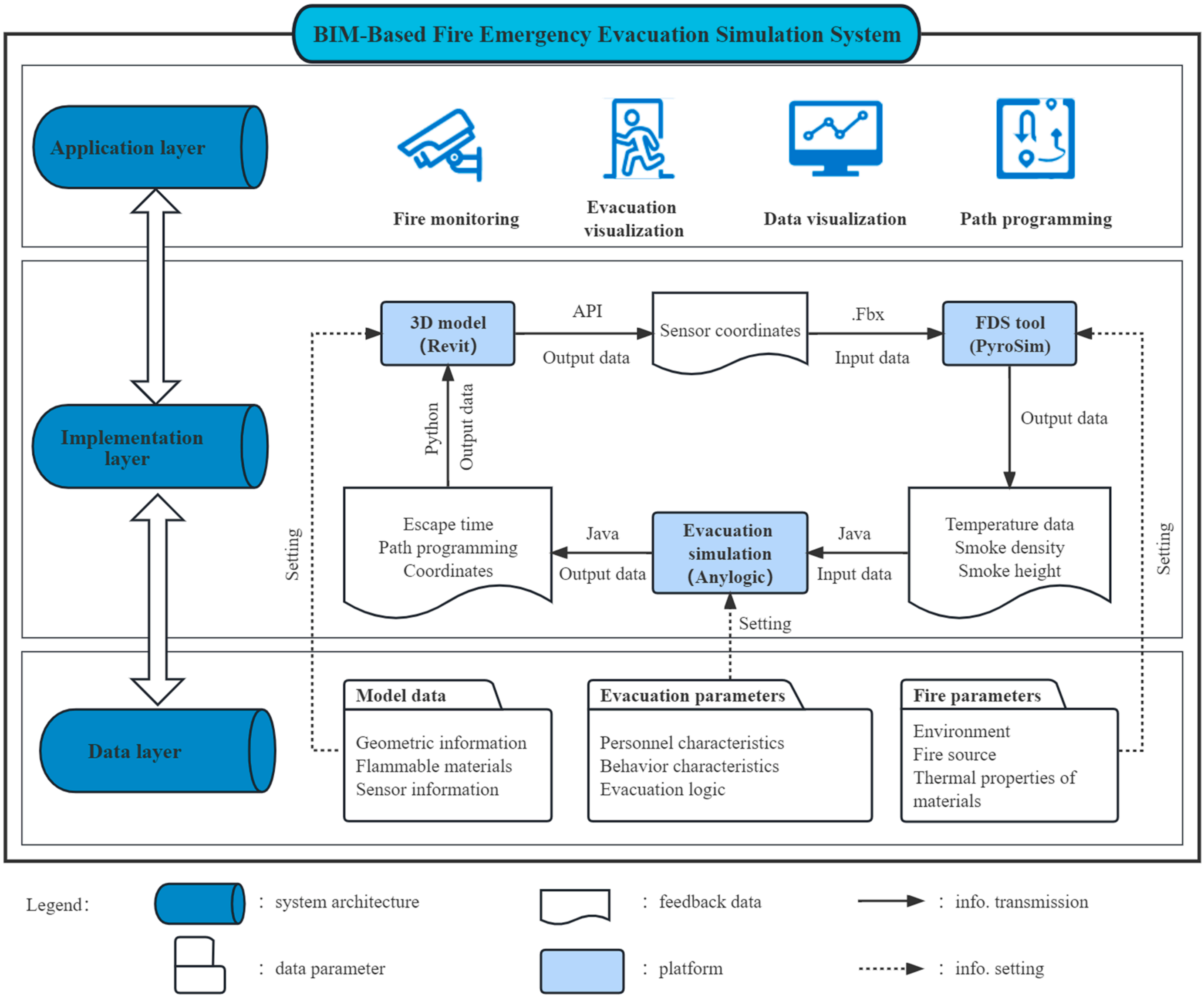


Fig. 2. Framework and operating mechanism of the fire emergency evacuation simulation system.

reflects real-life scenarios. In addition, similar to fire simulations, evacuation simulations require the integration of sensor data obtained from the actual environment or generated by the fire simulation. Only accurate sensor positioning coordinates can fulfill these functional requirements.

BIM data need to be imported into the AnyLogic simulation platform in the *.dae format.* To ensure the integrity and accuracy of information in the subsequent evacuation simulation, the API tool (SimLab Collada Exporter for Revit) developed by SimLab Soft is used, and the 3D view in the *.dae* format by selecting the appropriate zoom ratio and other settings. The information model in the AnyLogic platform uses pixels as the unit of measurement. The default exchange unit between pixel and meter is: 1 m = 10 pixels.

To correctly output sensor coordinates, sensor coordinates need extra processing between the Revit and AnyLogic platforms. The x axis is positive pointing from left to right in the coordinate systems of the Revit and AnyLogic platforms, and the y axis is positive pointing downwards in the AnyLogic platform and pointing upwards in the Revit system. Therefore, the transformation formula is shown in Eq. (1).

$$\begin{cases} x_{Anylogic} = x_{Revit} \\ y_{Anylogic} = -y_{Revit} \end{cases} \tag{1}$$

In addition, this research integrates evacuation routes into the evacuation simulation platform using the information model. The customized development function of Dynamo is used to connect the coordinates of the evacuation simulation output in sequence to visualize the evacuation path within the BIM platform. Firstly, the coordinates of evacuated individuals are read and then separated. These coordinates are merged to form the coordinates of each point. Then, a Python code is written using the Python script module to establish connections between batches of points and points in a sequence. This enables two-way interaction between the results of the evacuation simulation and the BIM platform.

4.4. Optimization of evacuation planning routes

If the evacuation simulation assumes that evacuees select the shortest evacuation route, then the evacuation of people at each exit is completed one after another using an algorithm. Based on the judgment of the shortest distance, some evacuation exits often cause congestion during the evacuation, while others are less efficiently used. The congestion of evacuation exits can easily exacerbate panic among evacuees, leading to behaviors such as turning back to find their way and causing fearful trampling [46,47]. At the same time, fire smoke

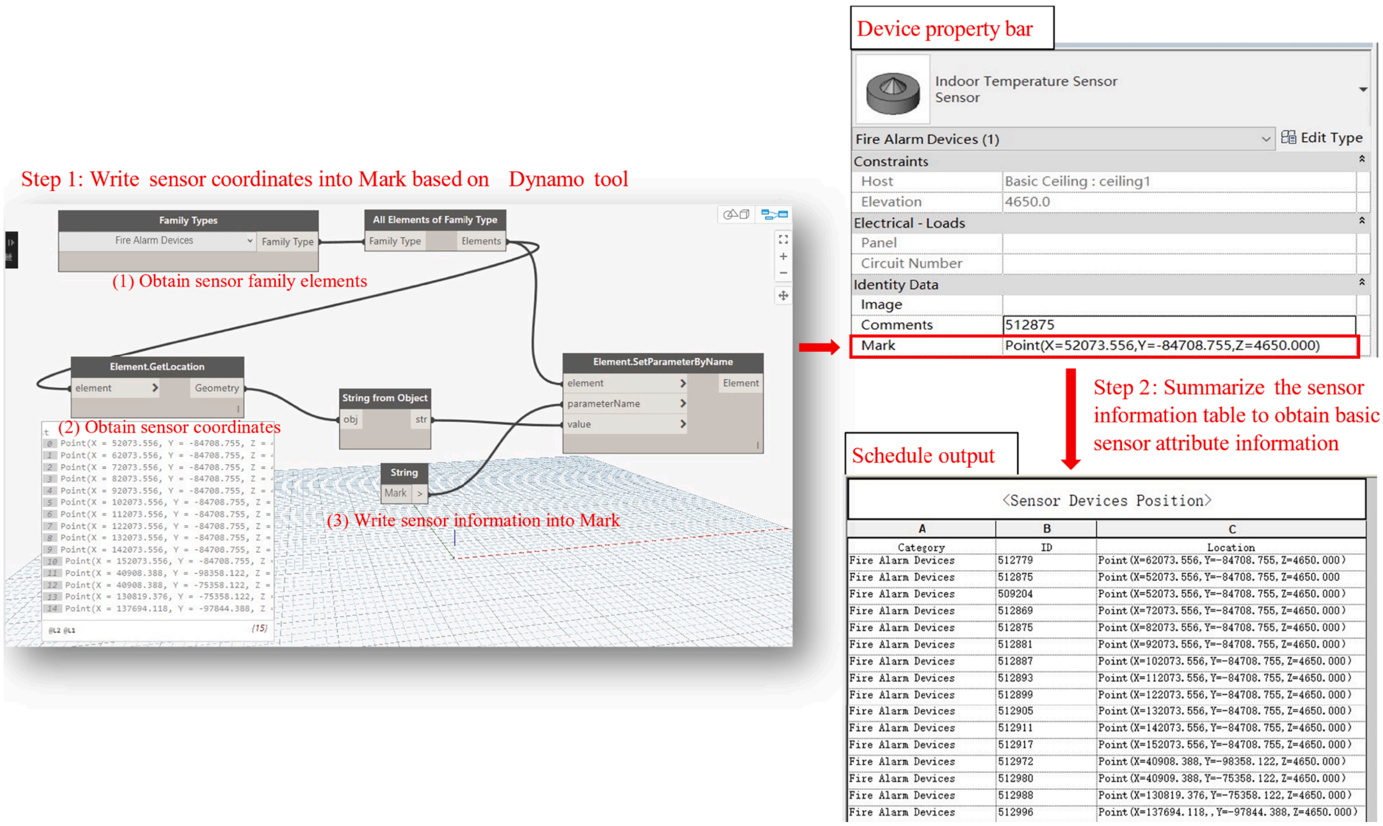


Fig. 3. Output of sensor coordinates using Dynamo.

continues to spread, which will pose a great threat to evacuees' behavior. Thus, this study proposes a route optimization algorithm. As shown in Table 2, the design logic of the algorithm is: (1) considering evacuation efficiency of all exits to set their evacuation area; (2) determining whether there is a corresponding evacuation area in the evacuees' coordinates; (3) judging the selection of evacuation exits; (4) considering the difference between the evacuation area of each exit and its total evacuation time, adjusting evacuation area for each exit, and iterating continuously; and (5) stopping iteration when the evacuation time difference of each exit is no more than 10 s, indicating that each evacuation exit has reached saturation at the same time. The highest evacuation efficiency is ensured.

4.5. Output of evacuation simulation results

The evacuation simulation result is crucial for the fire emergency evacuation system, as it directly impacts the evacuation strategies chosen by both operators and evacuees. The evacuation simulation in this study generates three types of outputs:

(1) Visual scenes. The visual output of the evacuation simulation can demonstrate the entire evacuation process. For example, Fig. 5 shows the visualization of an individual evacuation route during the evacuation process after the Java code is written in the AnyLogic platform;
(2) Dynamic charts. Charts can be obtained during the evacuation simulation process, which changes according to time. As shown in Fig. 6(a), users can assess the current level of congestion in the designated area based on the density graphs. As shown in Fig. 6(b), users can estimate the distribution of time required for different individuals to evacuate by analyzing the histograms of the number of people involved in the dynamic evacuation. As shown in Fig. 6(c), users can monitor the real-time fluctuations in the number of individuals presented in the designated area and the number of people who have been evacuated. These figures can assist operations staff of large infrastructures in making effective decisions regarding assistance; and
(3) Evacuation route coordinates. Analysis of the time-step statistics of evacuees' coordinates during the evacuation process can provide insights into the route selections of individual evacuees. In the event of a fire, an evacuation route guidance plan can be created for evacuees based on their locations to prevent them from passing through hazardous areas and causing unnecessary congestion.

5. Case study

5.1. Case description

To validate the proposed BIM-based fire emergency evacuation simulation system, this study used the Chegongmiao underground subway station located in Shenzhen, China as an example. This two-story station integrates commercial and transportation facilities and is one of the busiest subway stations in China. This station includes four subway lines (Line 1, Line 7, Line 9, and Line 11). This study focused on Line 11, which runs through the airport, railway station, and commercial area and is the busiest line. Therefore, this study constructed a 3D information model following the LoD 300 standard [48]. The model provides detailed representations of geometry, materials, and sensor coordinates.

5.2. Establishment of fire simulation

To establish a fire simulation model, the 3D model of the Chegongmiao station was exported in the *.fbx* file format and imported into the PyroSim software for model verification. Firstly, the model was

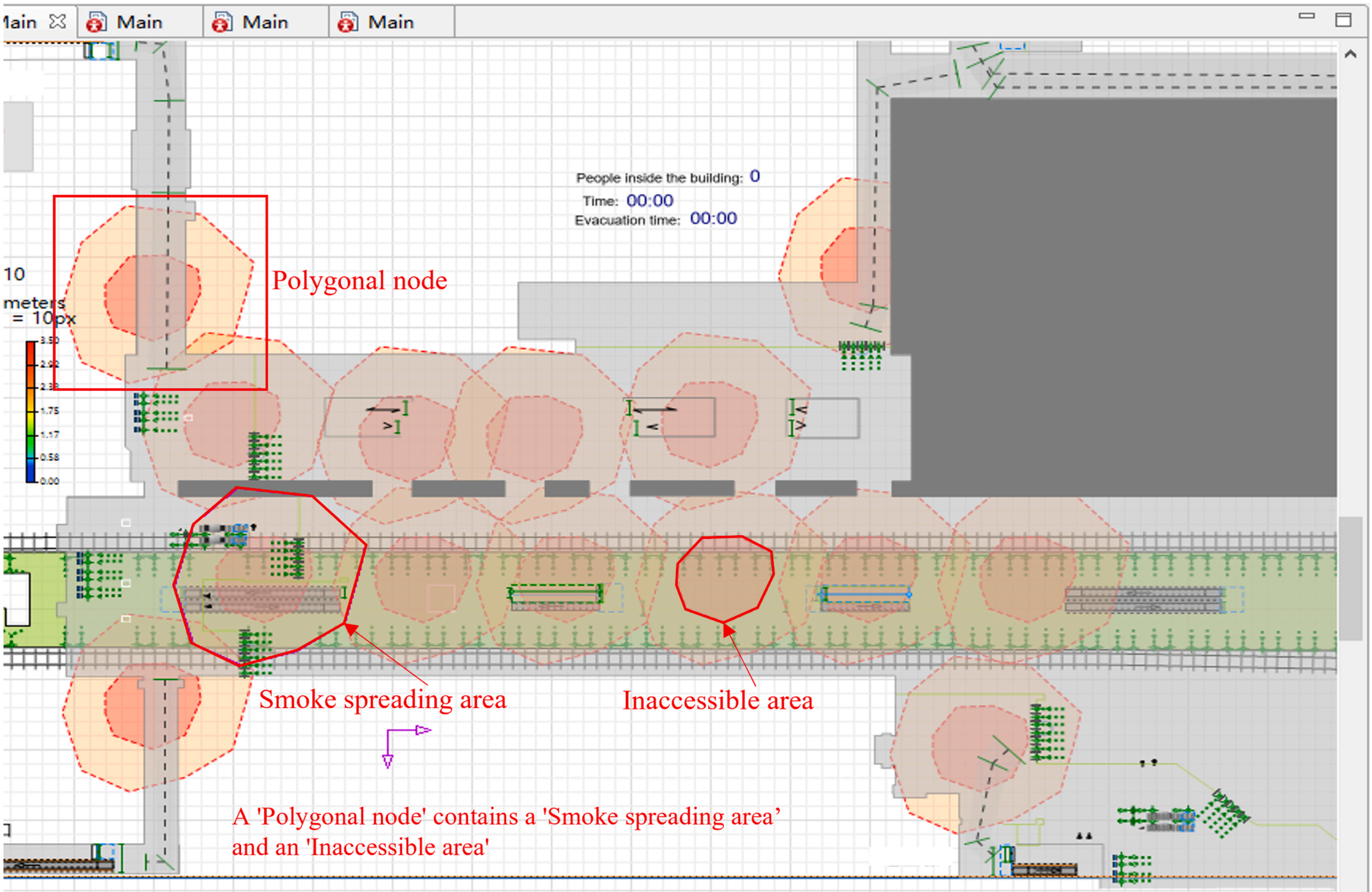


Fig. 4. Pedestrian speed control through sensor data.

Table 1
Algorithm for interaction of fire data with the evacuation simulation platform.

Notation representation description	
$T_{\text{evacuation}}$	Time of evacuation obtained in the evacuation simulation platform
T_{fire}	Time of fire
H	Smoke height
V	Pedestrian evacuation speed
CoE	Coordinates of evacuees
SA	Smoke area

Algorithm 1: Interaction between fire data and evacuation simulation platform
Input: T_{fire}, H;
Output: V;
(1) For each T_{fire};
(2) If (T_{fire}==T_{evacuate});
(3) If (H > 1.2 && H < =1.5 && CoE contain SA);
(4) $V = V \times \frac{H-1.2}{1.5-1.2} \times 90\%$;
(5) If (H > 1 && H < =1.2 && CoE contain SA);
(6) $V = V \times \frac{H-1}{1.2-1} \times 60\%$;
(7) If (H < =1 && CoE contain SA);
(8) Return 0;//Unable to walk, need to avoid;
(9) End for;

Table 2
Design logic of the route optimization algorithm.

Notation representation description	
EA_i	Initializing evacuation area for each exit
TET_i	Total evacuation time for each exit
CoE	Coordinates of evacuees
NEA_i	New evacuation area for each exit
$NTET_i$	New total evacuation time for each exit
ER	Evacuation route

Algorithm 2: Route optimization
Input: EA_i (determined by the shortest path algorithm), TET_i;
Output: NEA_i, $NTET_i$, ER
(1) If (TET_i < *other* TET_i);
(2) If (*other* $TET_i - TET_i$ > = 10);
(3) Define new evacuation area for each exit: EA_i.setX(EA_i.getX()+10); EA_i.setY (EA_i.getY()+10); // Move 10 px each in the X and Y directions
(4) If (CoE contain NEA_i);// The judgment of evacuated individuals on the evacuation route
(5) Return evacuation path corresponding to exit: ER
(6) Define new total evacuation time for each exit: $NTET_i$;
(7) For (*other* $NTET_i - NTET_i$ > = 10);
(8) Execute step (3)–(7);
(9) Else end;

evaluated to ensure that the combustibles included had corresponding definitions, to prevent significant discrepancies between simulation results and the real-world conditions. Then, the simulation model of the subway station was divided into a 3D network consisting of 426,020 (358 × 238 × 5) grids. After that, according to sensor data in the BIM database, identities and coordinates of temperature and smoke sensors were established in PyroSim, as shown in Fig. 7.

Regarding parameter settings, the initial ambient temperature, humidity, and wind speed in the Chegongmiao subway station were determined through field investigation and incorporated into the fire simulation model, as shown in Table 3. This study referred to the standard ISO/TS 16,733 Fire Safety Engineering and set the fire scenario with the most unfavorable fire spread speed to ensure that the result was more conservative and safer [49]. During the simulation, this study intentionally created the most unfavorable scenario by starting a fire at the security inspection location for Line 11 in the station hall during the

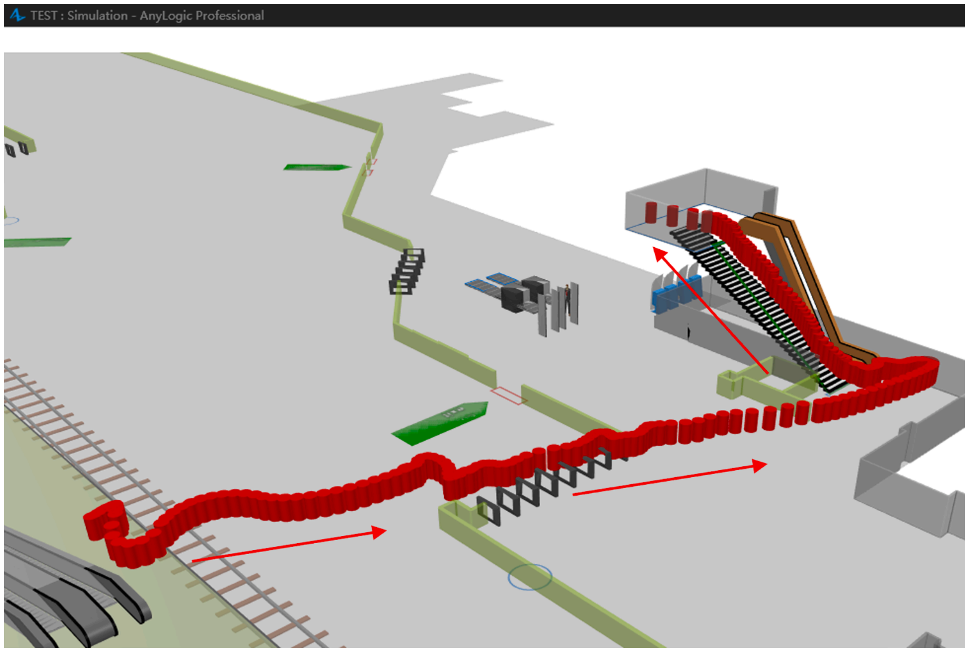
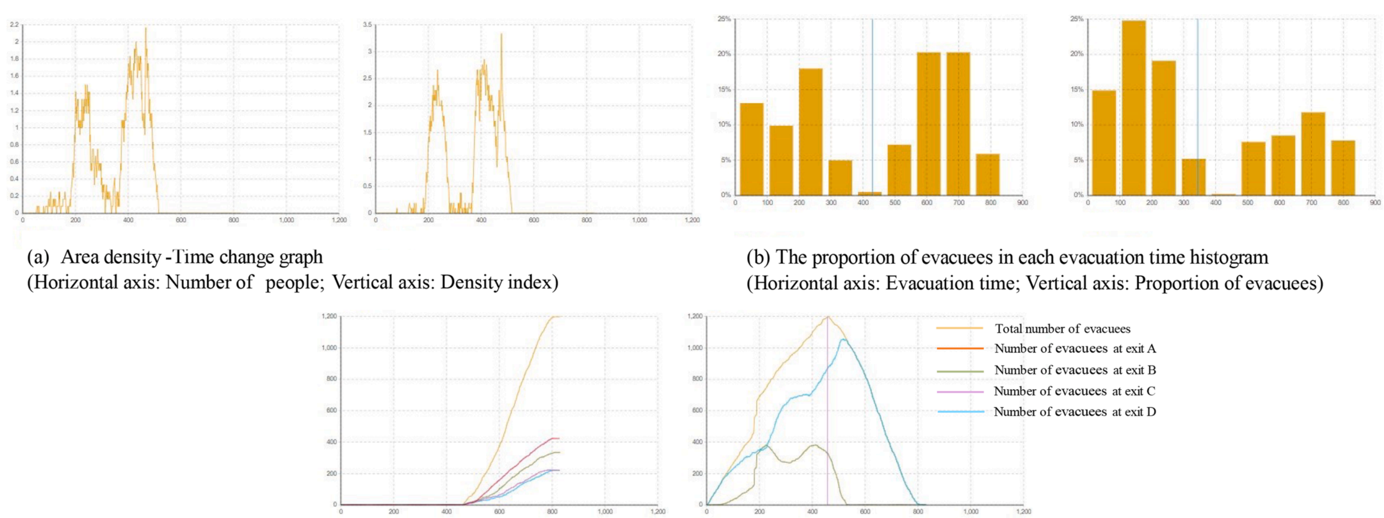


Fig. 5. Visual output of individual evacuation routes.



(a) Area density -Time change graph
(Horizontal axis: Number of people; Vertical axis: Density index)

(b) The proportion of evacuees in each evacuation time histogram
(Horizontal axis: Evacuation time; Vertical axis: Proportion of evacuees)

(c) Number of evacuees and people not evacuated -Time change graph
(Horizontal axis: Evacuation time; Vertical axis: Number of evacuees)

Fig. 6. Dynamic chart output.

morning peak period.

5.3. Establishment of evacuation simulation

The evacuation simulation model used the model in the *.dae* format output by the Revit API to establish the simulation environment and generate a 3D representation, as shown in Fig. 8. "Wall", "Rectangular Wall", and "Circular Wall" were used to represent blocking effects of walls, pillars, and railings in the information model on pedestrians; "Target Line" marked the starting point of pedestrians; "Service With Lines" and "Service With Area" demonstrated the function of facilities and equipment in the station, such as gates, security check, and ticket machine, representing the effect of queuing on a line or in the area; "Rectangular Node" represented the function of changing floors up and down by stairs; "Escalator Group" realized the expression of escalator function. Through the above definition, the expression of the evacuation simulation model in the station was basically completed.

In terms of evacuation parameters setting, this study combined field investigation and literature review to acquire parameters related to behavioral characteristics and pedestrian flow. The physical characteristics of evacuees can affect the speed of evacuation and, consequently, the decision-making process. Different age groups were selected to estimate the physical characteristics of shoulder width, chest circumference, and height of evacuees and the proportion of evacuees in Table 4. These selections were made based on suggestions from the literature [50,51] and surveys. During an evacuation, evacuees move at varying speeds in different locations. For example, evacuees generally evacuate slower on stairs than in passageways or on the ground. Therefore, this study determined the evacuation speed for different age groups in Table 5 as suggested in the literature [52,53]. The flow of people in the

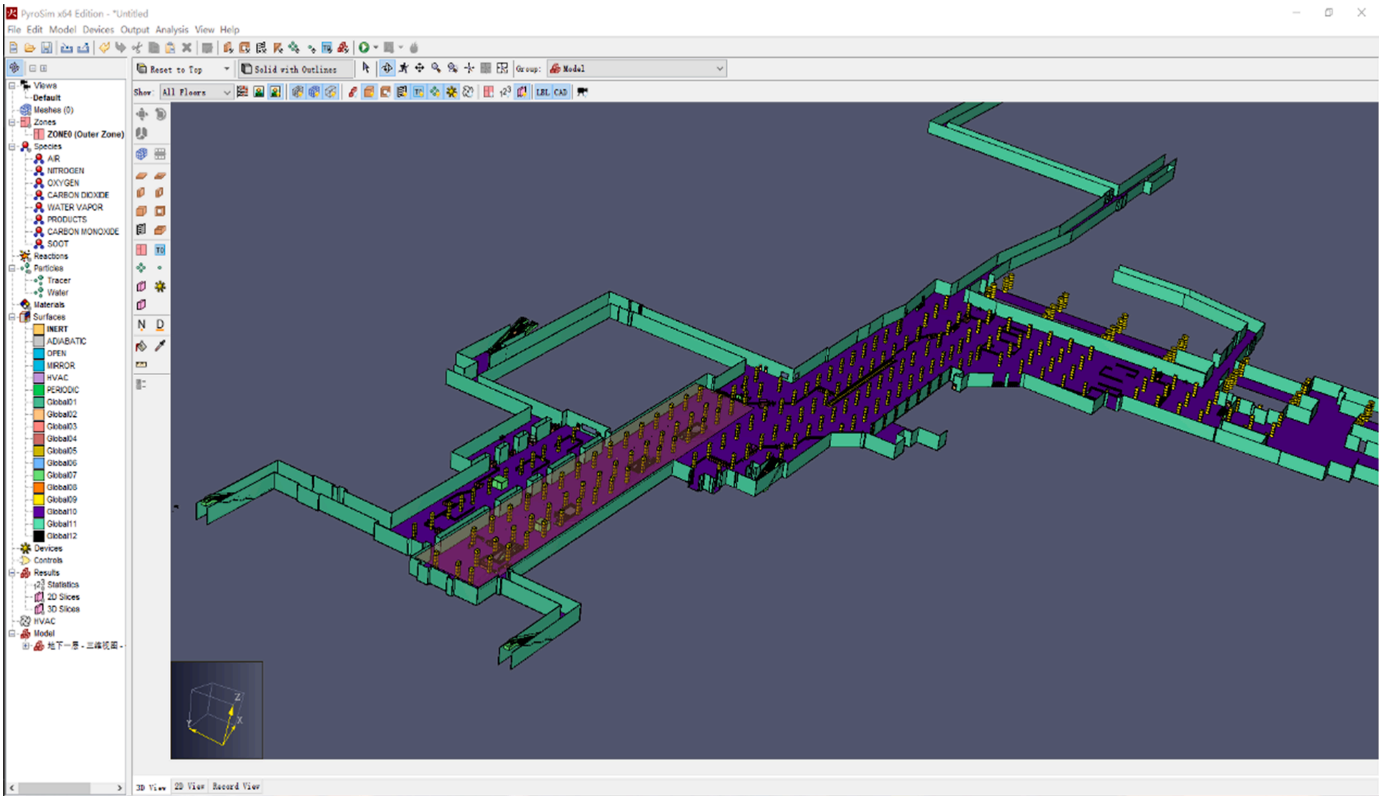


Fig. 7. Fire simulation model in PyroSim.

Table 3
Setting of initial conditions for fire simulation in the Chegongmiao subway station.

Initial conditions	Parameter value
Ambient temperature	25 °C
Ambient humidity	55%
Ambient wind speed	0 m/s
Simulation duration	1200 s
Fire growth coefficient	0.01876 kw/s^2
Type of fire growth	Super-fast
Release rate of fire heat	4 MW

Table 4
Characteristic parameters of evacuees.

Type of evacuees	Shoulder width (m)	Chest circumference (m)	height (m)	Proportion (%)
Adult male	0.50	0.93	1.70	34
Adult female	0.44	0.87	1.60	34
Elder	0.45	0.93	1.60	15
Children	0.35	0.61	1.20	17

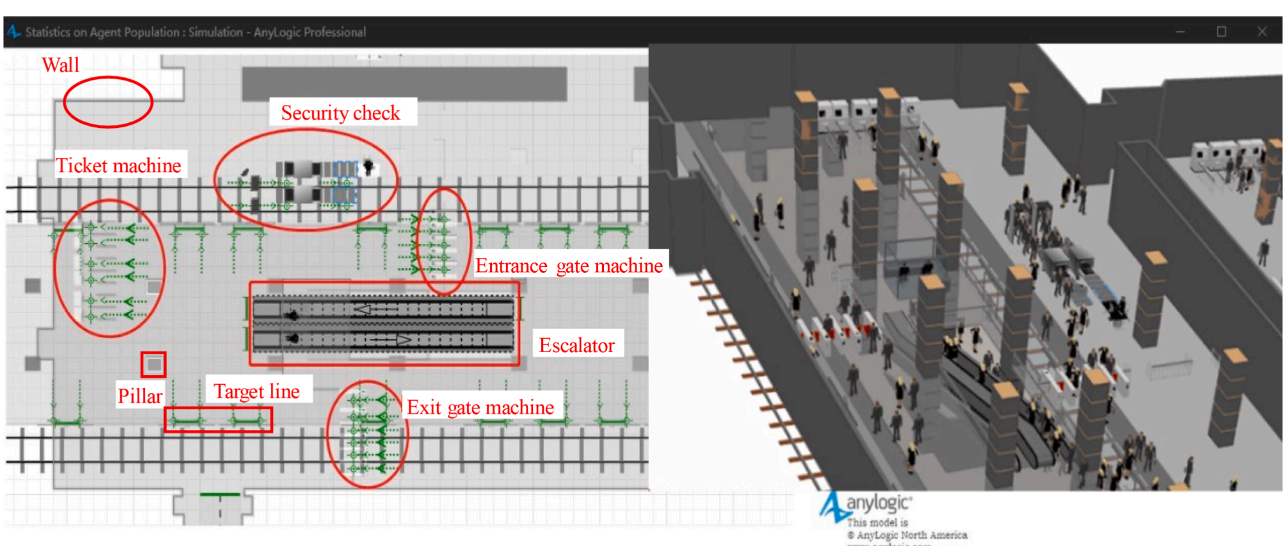


Fig. 8. Local expression of the environment in the Chegongmiao subway station.

Table 5
Walking speed of pedestrians at different locations.

Person	Aisle (m/s)	Going upstairs (m/s)	Going downstairs (m/s)	Ground (m/s)
Adult male	1.39	0.75	0.95	1.58
Adult female	1.22	0.66	0.83	1.43
Elder and children	1.06	0.52	0.40	1.17

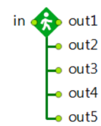
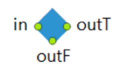
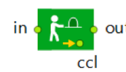
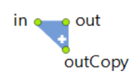
station is also an essential factor affecting evacuation. In this study, the flow statistics during the peak hours of working days were used as the flow parameters for the simulation, as shown in Table 6, so that the evacuation simulation could be carried out under the highest evacuation pressure, providing the most valuable simulation reference.

5.4. Setting of evacuation logic

The setting of evacuation logic is to invoke components of the evacuation simulation model using modules from the pedestrian library. This creates a logical framework including normal station entry and exit procedures and evacuation procedure. The modules are connected sequentially based on discrete events. The logic model mainly includes the following modules: pedestrian source module, layer change module, pedestrian selection output module, pedestrian service module, and delay module, as shown in Table 7.

Pedestrian logic design mainly included four processes: entering the station, transfer, leaving the station, and evacuation, as shown in Fig. 9. The logic design of each process was based on the actual scene of the corresponding process to ensure that the simulation results provide valuable guidance for actual evacuation [54,55].

Table 7
Introduction of each module in the logic model.

Module name	Icon	Function
pedSource		Pedestrian source module: it is usually originated as a pedestrian flow, and can customize walking speed and flow of pedestrians according to needs.
pedChangeLevel		Changing layer module: it moves the pedestrian flow to a new floor, and if combined with the inclined rectangular node and the target line, can be used to express stairs.
pedSelectOutput		Pedestrians' selection output module: pedestrians select outlets (5) that meet predefined probability or conditions.
selectOutput		Selection output module: an agent selects outlets (2) that satisfy predefined probability or conditions.
pedService		Pedestrian service module: if combined with line service in modeling, regional service, and other functions, it can express security check, ticket purchase, going through gates, platform queuing, and so on.
split		Splitting module: it creates other agents for an incoming agent, which is used during simulation to express pedestrians separating bags for security inspection.
delay		Delay module: it ensures an entering agent is delayed and waited in the original position. In the simulation, this module is used to express waiting of security check and going through gates.

6. Results and validation

6.1. Application of the fire emergency evacuation simulation system

With the use of the fire emergency evacuation simulation system, fire simulation was executed on the selected position. The fire simulation result file was viewed using Smokeview, which is a function of PyroSim. The visual animation depicting the growth and spread of the fire in the station hall was viewed. As shown in Fig. 10, at 300 s, the smoke height at exits B and C of the Chegongmiao station was less than 1.0 m, and the smoke infiltrated exits A and D and dispersed to the station hall, reaching a height of 1.5 m; at 547 s, the smoke height at exits A and D was less than 1.0 m. Evacuees must complete evacuation before the smoke reaches a height of 1.0 m [56]. Based on the analysis, it was concluded that exits B and C were out of service at 300 s, with an available evacuation time of 300 s, and the available evacuation time for exits A and D was 547 s.

The sensor data for temperature, smoke density, and smoke height collected during the fire simulation were input into the AnyLogic platform through its customized development for evacuation simulation. This evacuation simulation platform sequentially reads the sensor data according to the simulation time schedule. This study simulated a fire occurring at the security checkpoint during the morning peak period (the flow of people in the station reached 1200 people). It was assumed that the total time of evacuees from detection of the fire or fire alarm to start of evacuation is 20 s, considering their psychological and behavioral characteristics. When the temperature exceeded 60℃, the platform simulated pedestrians evading the fire. When the smoke height reached 1.5 m or 1.0 m, pedestrian speed in the restricted area decreased to 90% or 60% of their normal walking speed. The platform also simulated the impact of evacuees' bowing and bending behaviors to avoid heavy smoke on their walking speed [57].

Table 6
Peak pedestrian flow.

Pedestrian evacuation range	Composition of pedestrian evacuees	Data sources	Flow of people
Subway platform	Passengers getting off	Field investigation	3–5 per door
Station hall	Passengers entering the station	Automated fare collection system	16,303
	Passengers leaving the station		19,248
	Passengers entering the station (Line 11)		3085
	Passengers leaving the station on (Line 11)		3643
Transfer channel	Passenger flow in both directions of Line 11	Field investigation	12,010

There are four exits in the subway station. The evacuation area of each exit in the initial simulation is shown in Fig. 11. Different colors represent the areas of different exits based on the approach of the shortest evacuation route.

According to the evacuation time theory, if the available safe escape time (ASET) is longer than the required safe escape time (RSET), evacuees can be safely evacuated [58]. The final evacuation result is shown in Fig. 12. In addition, the density of evacuees during the evacuation process is presented in Fig. 13, and the peak density value was set as the number of people per unit area reaching 3.5.

6.2. Evacuation route optimization and output

According to the preliminary evacuation simulation results using the shortest route approach, the evacuation efficiency of each exit was judged. Exits with higher evacuation efficiency should be assigned larger evacuation areas. It can be seen from Figs. 12, 13 that, exits D and A had a large density of evacuees and were prone to congestion, and exit D completed evacuation much later than other exits. Therefore, it was

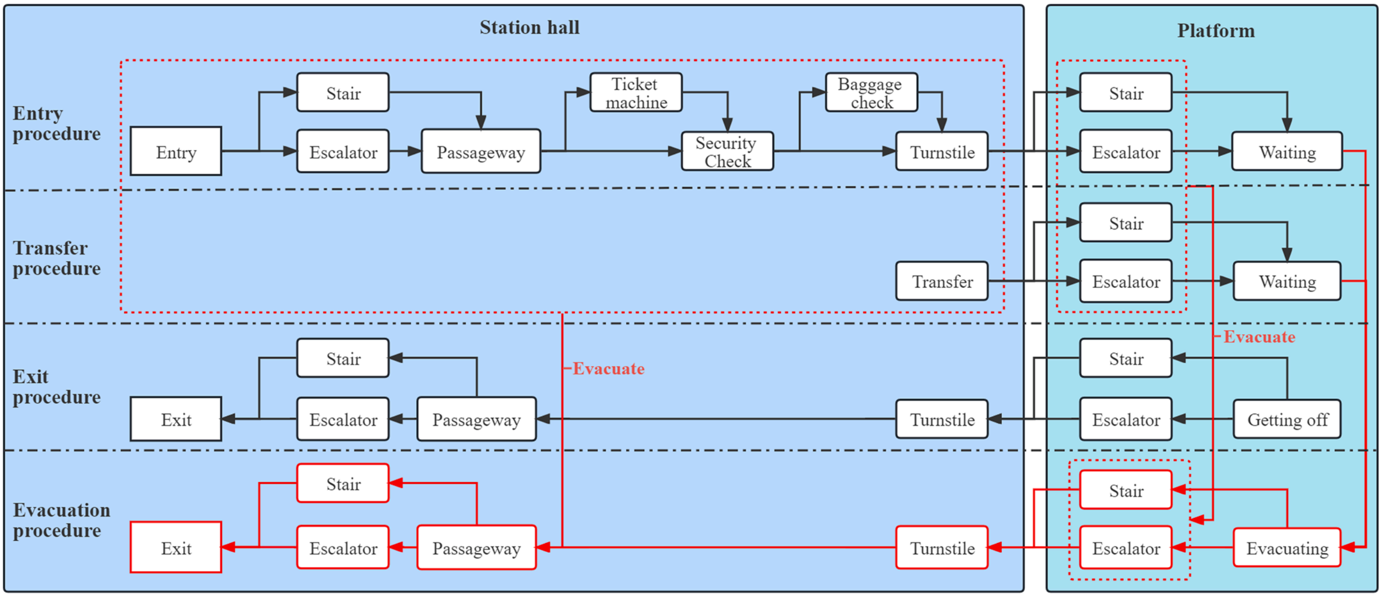


Fig. 9. Pedestrian flow logic (black route for regular subway travel and red route for evacuation).

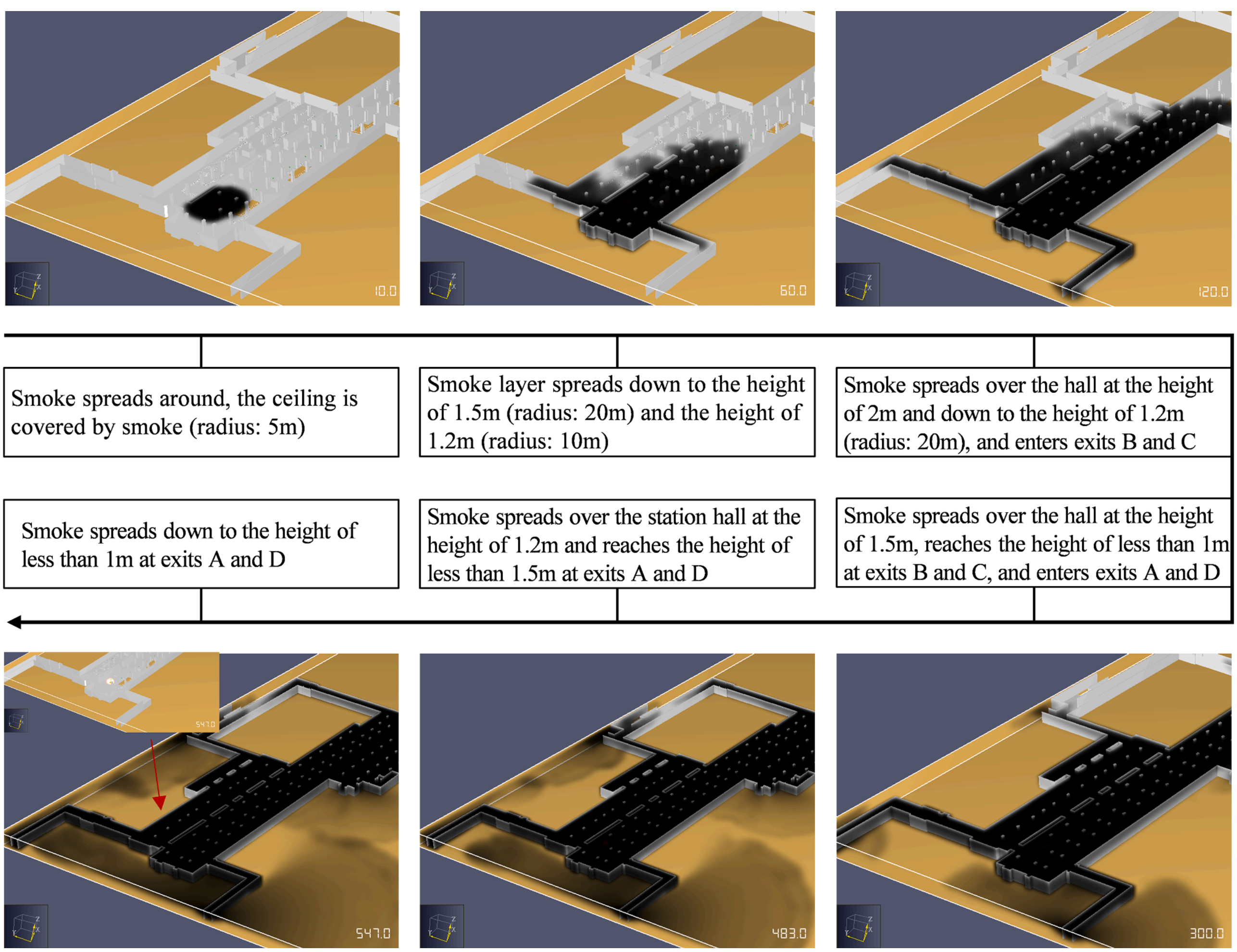


Fig. 10. Smoke spread process during fire simulation in the Chegongmiao subway station.

determined that exits B and C should undertake more evacuation areas, while exits D and A should be allocated less evacuation areas.

To improve the overall efficiency of evacuation, this research designed an optimization algorithm aiming at achieving equal evacuation time at each exit of the subway station. As shown in Table 2, each simulation result was compared with the previous one to automatically

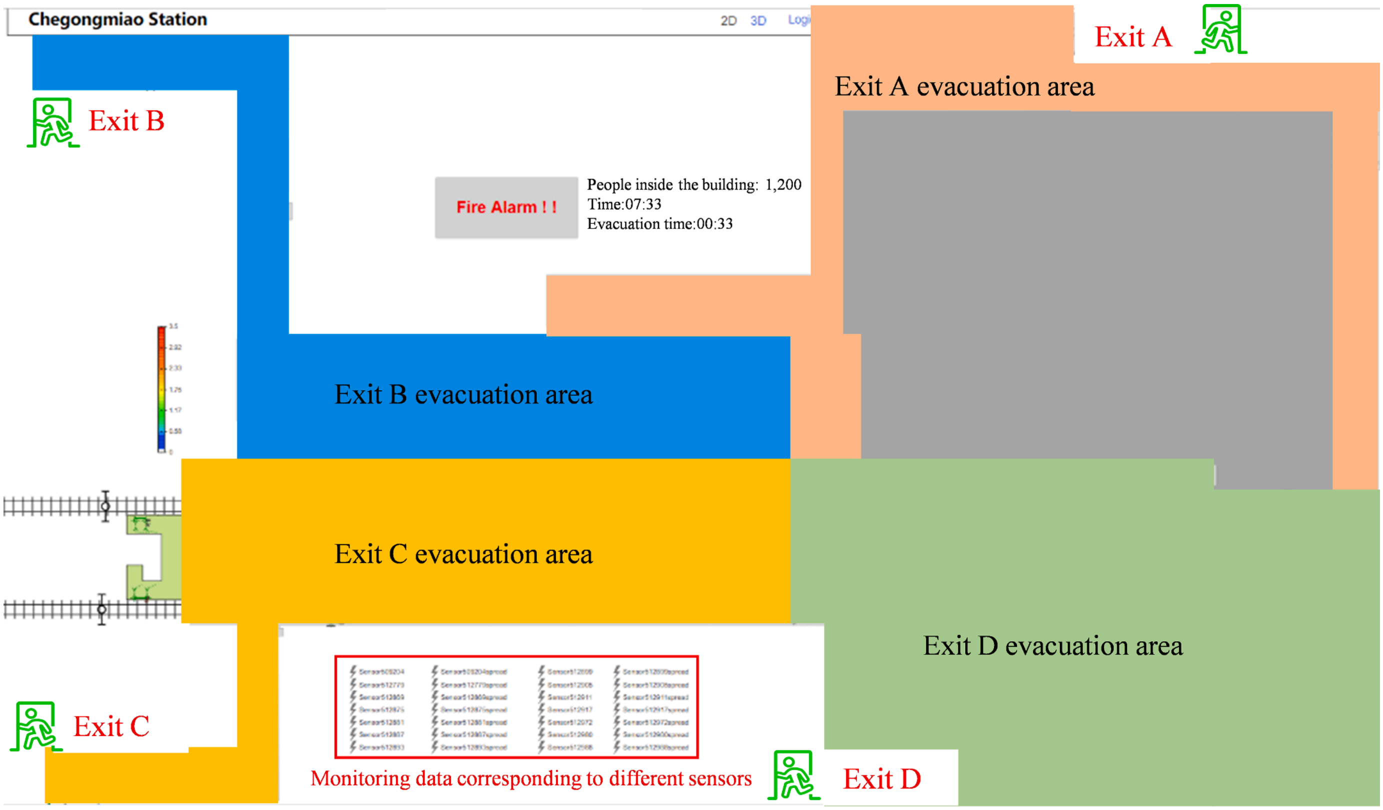


Fig. 11. Initialization of the evacuation area for each evacuation exit.

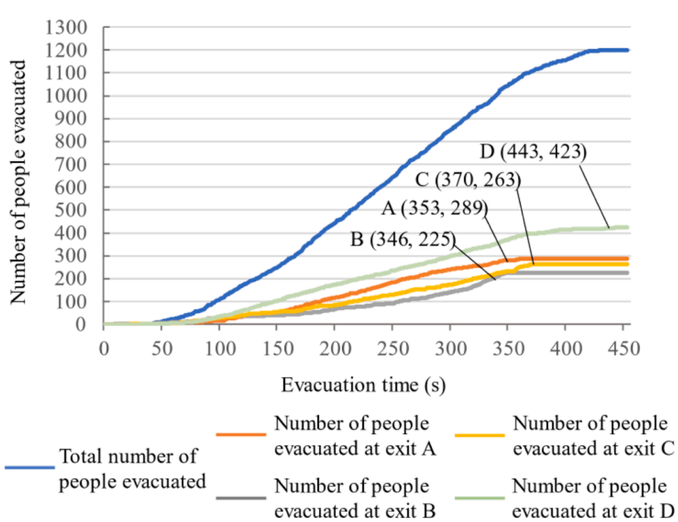


Fig. 12. Initial fire evacuation results of the Chegongmiao subway station.

adjust the size of each evacuation area through iterative simulation optimization. When the time difference for evacuation at each exit is 10 s or less, the iteration will not proceed. The optimal evacuation is reached.

The results of the evacuation simulation based on the regional route selection algorithm are shown in Fig. 14. Through optimizing the algorithm, the final evacuation time (386 s) was reduced by 57 s, compared to the traditional evacuation algorithm (443 s), as shown in Table 8. Thus, the evacuation efficiency improved by 12.87%. This result was comparable to latest studies. For example, a virtual reality crowd evacuation simulation method based on an improved social force model was proposed, which improved evacuation efficiency by 12.53% [59]. Besides, an overall optimization degree of evacuation route of 12.8% was achieved using multi-objective optimization algorithm [60]. Moreover, a stochastic user equilibrium model was used to reduce evacuation time by 15% at the Qingdao May 4th Square subway station in China [61]. Therefore, the method proposed in this study has well optimized evacuation efficiency. The evacuation time of exits A (386 s) and D (378 s) was basically controlled to the same amount, and the evacuation time difference between exit B (287 s) and exit C (296 s) was also less than 10 s. Since the fire in this case study was set at the security checkpoint, which is located close to exits B and C of the subway station, smoke filled these two exits, making it soon impossible for people to evacuate. Thus, the evacuation time of exits A and D was longer than that of exits B and C. It was concluded that the system and optimization algorithm proposed in the present study are applicable and reliable.

Finally, Dynamo was used to integrate the resulting evacuation route with the 3D information model of the subway station. This tool collected and connected the coordinates of the evacuation simulation output in sequence for visualizing evacuation routes in the BIM platform. Fig. 15 shows the route of an evacuated individual to reach the station hall floor, either by using the stairs or the escalator. The BIM platform reads individual evacuation coordinates and connects them to represent the evacuation route in the information model. Therefore, the operations and maintenance team can provide personalized evacuation route guidance for people in the station based on the information model. This significantly enhances the efficiency of evacuation decision-making.

7. Conclusions

Efficient and safe evacuation systems play a crucial role in protecting people's lives during major fires in large public infrastructures. This study developed a simulation system for fire emergency evacuation. The system facilitates data interaction between infrastructure, fire, and evacuation, taking into account the impact of real-time fire conditions and evacuation behavior on the evacuation to optimize escape routes for

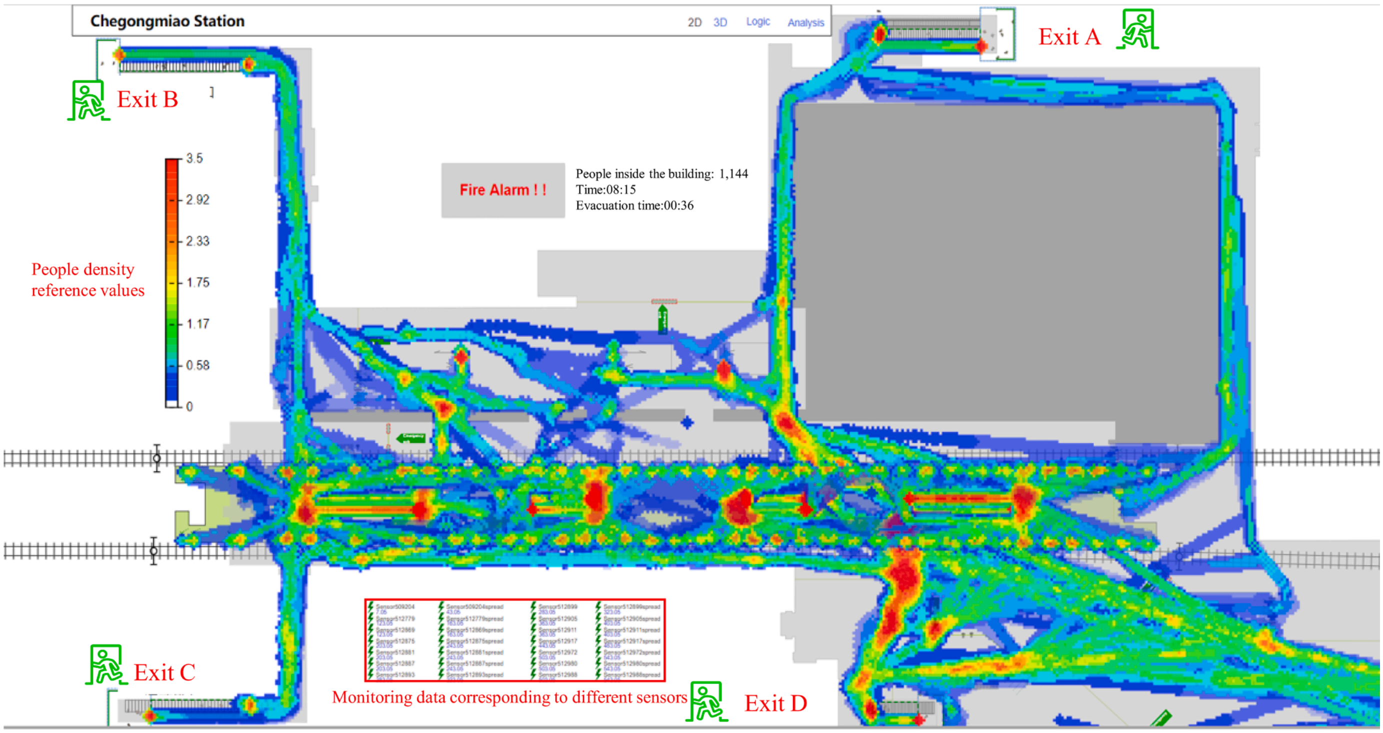


Fig. 13. Visualization of crowd density during the evacuation.

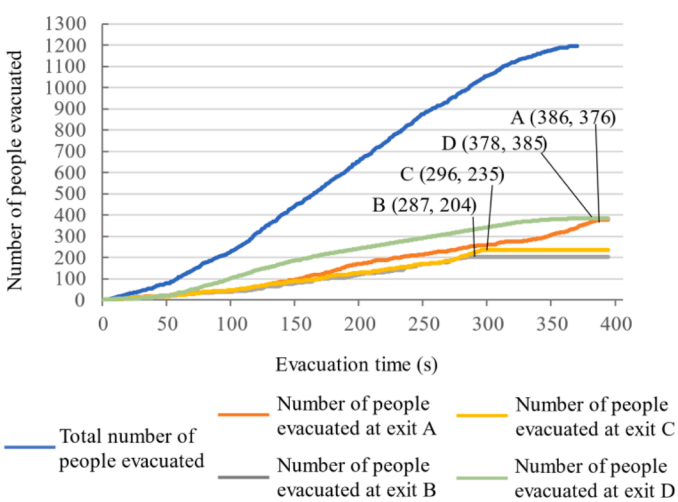


Fig. 14. Fire evacuation results after route optimization.

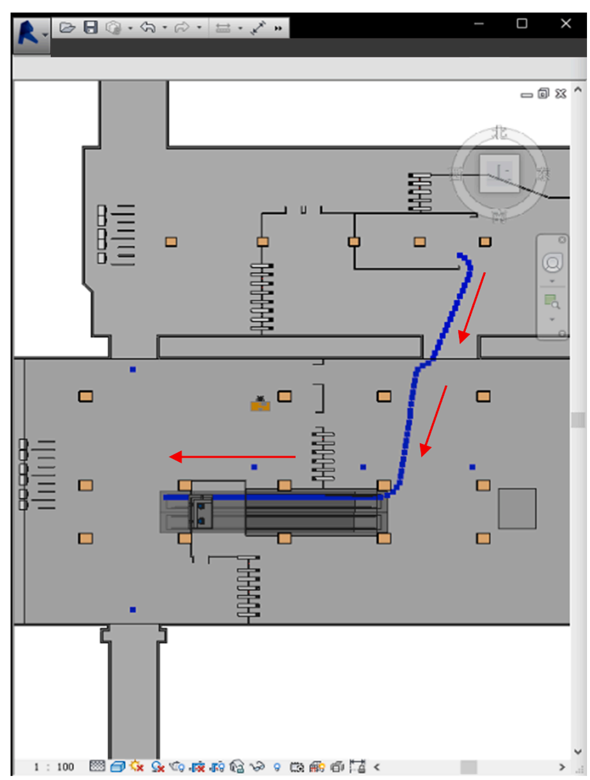


Fig. 15. Display of the evacuation route in the BIM platform.

Table 8
Comparison of two evacuation strategies.

Exit	T_{ASET}/s	T_{REST}/s Original → Optimized	Number of evacuees Original → Optimized	Evacuation
A	547	353 → 386	289 → 376	Safety
B	300	346 → 287	225 → 204	Safety
C	300	370 → 296	263 → 235	Safety
D	547	443 → 378	423 → 385	Safety

evacuees. Firstly, a realistic 3D information model is created in Revit, and the information and sensors are imported into the PyroSim platform for fire simulation. The fire data is then transferred to the AnyLogic simulation platform to reflect the impact of the fire on evacuation. In the evacuation simulation, evacuation parameters of different age groups are set for simulation based on their physical characteristics and behavioral performance to ensure the authenticity of the simulation results. To achieve the best evacuation efficiency and provide personalized evacuation routes for evacuees, this research designed an evacuation route optimization algorithm considering the congestion level of each exit and the location of each evacuee. Additionally, the system visualizes the escape routes of individuals.

This study took a large underground subway station in Shenzhen, China as an example for fire emergency and evacuation simulation to validate the proposed system. The results indicated that the system can assist in planning the evacuation route and calculating the optimal escape route according to the locations of evacuees and available exits to ensure safety and minimize casualties. Finally, the system significantly shortened the evacuation time and improved the overall efficiency of the evacuation by 12.87%. Individual evacuation routes can be visually displayed on the BIM platform.

The main contributions of this study include:

(1) Developing a fire emergency evacuation simulation system for infrastructure management by integrating BIM, FDS, and computer simulation technologies. The social force model and agent-based model are combined to consider the influence of fire and evacuees' behavioral characteristics on evacuation, and visualize evacuation route guidance, which helps users make prompt decisions;
(2) Achieving the integration and interoperability of BIM data, fire data, and evacuation data from different platforms through visual programming technology, Java language, Python, and other technical means. This enhances intelligent fire emergency management, compared with previous fire emergency evacuation simulation systems; and
(3) Proposing the concept of overall evacuation efficiency and developing an evacuation path optimization algorithm that avoids congested routes based on the location of evacuees and improves the evacuation efficiency of each exit.

Despite the achievement of the research objectives, there are limitations. First, this study assumes a uniform reaction time of 10 s for all evacuees, without considering any variation in reaction time based on their locations. Second, considering that a fire may cause elevators to malfunction, evacuees are advised not to use elevators to escape. Future research would tailor the response time of each evacuee to further improve the realism of the evacuation simulation. Also, it is necessary to combine technologies such as indoor navigation and augmented reality to provide evacuees with a more accurate visualization of their evacuation routes.

CRediT authorship contribution statement

Zhikun Ding: Conceptualization, Methodology, Writing – original draft, Supervision, Funding acquisition. **Shengqu Xu:** Formal analysis, Investigation, Data curation, Visualization. **Xiaofeng Xie:** Resources, Project administration. **Kairui Zheng:** Software, Validation, Investigation, Data curation. **Daochu Wang:** Resources. **Jianhao Fan:** Project administration. **Hong Li:** Investigation. **Longhui Liao:** Conceptualization, Methodology, Writing – review & editing, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This research was conducted with the support of the National Natural Science Foundation of China (Grant no. 71974132), the Shenzhen Government Natural Science Foundation (Grant no. JCYJ20190808115809385), the Shenzhen Natural Science Fund (the Stable Support Plan Program nos. 20220810160221001 and 20220810155553002), and the Program of Shenzhen Key Laboratory of Green, Efficient and Intelligent Construction of Underground Metro Station (Program no. ZDSYS20200923105200001).

References

[1] Zverovich V, Mahdjoubi L, Boguslawski P, Fadli F, Barki H. Emergency response in complex buildings: automated selection of safest and balanced routes. Computer-Aided Civ Infrastruct Eng 2016;31(8):617–32.
[2] Himoto K, Suzuki K. Computational framework for assessing the fire resilience of buildings using the multi-layer zone model. Reliab Eng Syst Saf 2021;216:108023.
[3] Wang N, Xu Y, Wang S. Interpretable boosting tree ensemble method for multisource building fire loss prediction. Reliab Eng Syst Saf 2022;225:108587.
[4] Zhang D, Santos VH, Lofink BJ, Becher EM, Partyka A, Peng X, Sun L. Self-assembled intumescent flame retardant coatings: influence of pH on the flammability of cotton fabrics. Eng Sci 2020;12(2):106–12.
[5] Shi C, Zhong M, Nong X, He L, Shi J, Feng G. Modeling and safety strategy of passenger evacuation in a metro station in China. Saf Sci 2012;50(5):1319–32.
[6] Chan EYY, Huang Z, Hung KKC, Chan GKW, Lam HCY, Lo ESK, Yeung MPS. Health emergency disaster risk management of public transport systems: a population-based study after the 2017 subway fire in Hong Kong, China. Int J Environ Res Public Health 2019;16(2):228.
[7] Hu ZZ, Tian PL, Li SW, Zhang JP. BIM-based integrated delivery technologies for intelligent mep management in the operation and maintenance phase. Adv Eng Softw 2018;115:1–16.
[8] Gao X, Pishdad-Bozorgi P. BIM-enabled facilities operation and maintenance: a review. Adv Eng Inform 2019;39:227–47.
[9] Chen XS, Liu CC, Wu IC. A BIM-based visualization and warning system for fire rescue. Adv Eng Inform 2018;37:42–53.
[10] Zhao X, Yao J, Deng W, Ding P, Ding Y, Jia M, Liu Z. Intelligent fault diagnosis of gearbox under variable working conditions with adaptive intraclass and interclass convolutional neural network. IEEE Trans Neural Netw Learn Syst 2023;34(9):6339–53.
[11] Wehbe R, Shahrour I. A BIM-based smart system for fire evacuation. Future Internet 2021;13(9):221.
[12] Li Y, Lin X, Feng X, Wang C, Li J. Life safety evacuation for cross interchange subway station fire. Procedia Eng 2012;45:741–7.
[13] Pan ZH, Wei QC, Torp O, Lau A. Influence of evacuation walkway design parameters on passenger evacuation time along elevated rail transit lines using a multi-agent simulation. Sustainability 2019;11:17.
[14] Zhu X, Zhao X, Yao J, Deng W, Shao H, Liu Z. Adaptive multiscale convolution manifold embedding networks for intelligent fault diagnosis of servo motor-cylindrical rolling bearing under variable working conditions. IEEE/ASME Trans Mechatron 2023.
[15] Li X, Shao H, Lu S, Xiang J, Cai B. Highly efficient fault diagnosis of rotating machinery under time-varying speeds using LSISMM and small infrared thermal images. IEEE Trans Syst, Man, Cybern:Systems 2022;52(12):7328–40.
[16] Yang Y, Du HB, Yao G. A scientometric research on applications and advances of fire safety evacuation in buildings. Fire-Switzerland 2023;6(3):83.
[17] Nakaya N, Nemoto H, Yi C, Sato A, Shingu K, Shoji T, Sato S, Tsuchiya N, Nakamura T, Narita A, Kogure M, Sugawara Y, Yu Z, Gunawansa N, Kuriyama S, Murao O, Sato T, Imamura F, Tsuji I, Hozawa A, Tomita H. Effect of tsunami drill experience on evacuation behavior after the onset of the great east Japan earthquake. Int J Disaster Risk Reduct 2018;28:206–13.
[18] Kitazawa K, Hale SA. Social media and early warning systems for natural disasters: a case study of Typhoon Etau in Japan. Int J Disaster Risk Reduct 2021;52:101926.
[19] Kinateder M, Comunale B, Warren WH. Exit choice in an emergency evacuation scenario is influenced by exit familiarity and neighbor behavior. Saf Sci 2018;106:170–5.
[20] Xiao Y, Xu J, Chraibi M, Zhang J, Gou C. A generalized trajectories-based evaluation approach for pedestrian evacuation models. Saf Sci 2022;147:105574–82.
[21] Liu Q, He R, Zhang L. Simulation-based multi-objective optimization for enhanced safety of fire emergency response in metro stations. Reliab Eng Syst Saf 2022;228:108820.
[22] Zhao M, Fu X, Zhang Y, Meng L, Tang B. Highly imbalanced fault diagnosis of mechanical systems based on wavelet packet distortion and convolutional neural networks. Adv Eng Inform 2022;51:101535.
[23] Zhao X, Yao J, Deng W, Ding P, Zhuang J, Liu Z. Multiscale deep graph convolutional networks for intelligent fault diagnosis of rotor-bearing system under fluctuating working conditions. IEEE Trans Ind Inform 2023;19(1):166–76.
[24] Tang Y, Xia N, Lu Y, Varga L, Li Q, Chen G, Luo J. BIM-based safety design for emergency evacuation of metro stations. Automa Constr 2021;123:103511.
[25] Bina K, Moghadas N. BIM-ABM simulation for emergency evacuation from conference hall, considering gender segregation and architectural design. Archit Eng Des Manag 2021;17(5–6):361–75.
[26] Jeong CS. Development of a flood disaster evacuation map using two-dimensional flood analysis and BIM technology. Korean Soc Disaster Secur 2020;13(2):53–63.
[27] Liu Z, Li Y, Zhang Z, Yu W. A new evacuation accessibility analysis approach based on spatial information. Reliab Eng Syst Saf 2022;222:108395.
[28] Helbing D, Molnar P. Social force model for pedestrian dynamics. Phys Rev E 1995;51(5):4282–6.

[29] Wagner N, Agrawal V. An agent-based simulation system for concert venue crowd evacuation modeling in the presence of a fire disaster. Expert Syst Appl 2014;41(6): 2807–15.
[30] Blue VJ, Adler JL. Emergent fundamental pedestrian flows from cellular automata microsimulation. Transp Res Rec 1998;1644(1):29–36.
[31] Liu LF, Zhang HJ, Shi J, Geng JN. Evacuation simulation of a passenger ship fire based on a modified cellular automaton. Adv Theory Simul 2023;6(7):2300141.
[32] Berger C, Mahdavi A. Review of current trends in agent-based modeling of building occupants for energy and indoor-environmental performance analysis. Build Environ 2020;173:106726.
[33] Tsukahara M, Koshiba Y, Ohtani H. Effectiveness of downward evacuation in a large-scale subway fire using fire dynamics simulator. Tunn Undergr Space Technol 2011;26(4):573–81.
[34] Pei QY, Hao LJ, Chen CH, Zheng XL, He T. Minimum collective dose based optimal evacuation path-planning method under nuclear accidents. Ann Nucl Energy 2020; 147:107644.
[35] Du Y, Wu Q, Yao Y, Zhao Y, Zhang X, Hao Z, Xu H, Du Z. A cloud-based service for emergency evacuation of mine water inrush accidents. Concurr Comput 2021;33 (9):e6125.
[36] Deng K, Zhang Q, Zhang H, Xiao P, Chen J. Optimal emergency evacuation route planning model based on fire prediction data. Mathematics 2022;10(17):3146.
[37] Wang P, Zhang T, Xiao Y. Emergency evacuation path planning of passenger ship based on cellular ant optimization model. J Shanghai Jiaotong University (Science) 2020;25(6):721–6.
[38] Khakzad N. A methodology based on Dijkstra's algorithm and mathematical programming for optimal evacuation in process plants in the event of major tank fires. Reliab Eng Syst Saf 2023;236:109291.
[39] Guo K, Zhang L. Adaptive multi-objective optimization for emergency evacuation at metro stations. Reliab Eng Syst Saf 2022;219:108210.
[40] Rahman SAFSA, Maulud KNA, Pradhan B, Mustorpha SNAS, Ani AIC. Impact of evacuation design parameter on users' evacuation time using a multi-agent simulation. Ain Shams Eng J 2020;12(2):2355–69.
[41] Zhao Y, Liu H, Gao K. An evacuation simulation method based on an improved artificial bee colony algorithm and a social force model. Appl Intell 2021;51(1): 100–23.
[42] Jiang A, Mo Y, Kalasapudi VS. Status quo and challenges and future development of fire emergency evacuation research and application in built environment. J Inf Technol Constr (ITcon) 2022;27(38):781–801.
[43] Long XF, Zhang XQ, Lou B. Numerical simulation of dormitory building fire and personnel escape based on pyrosim and pathfinder. J Chin Inst Eng 2017;40(3): 257–66.
[44] Desogus G, Quaquero E, Rubiu G, Gatto G, Perra C. BIM and IoT sensors integration: a framework for consumption and indoor conditions data monitoring of existing buildings. Sustainability 2021;13(8):4496.
[45] Antonova VM, Grechishkina NA, Kuznetsov NA. Analysis of the modeling results for passenger traffic at an underground station using AnyLogic. J Commun Technol Electron 2020;65(6):712–5.
[46] Kuligowski E. Predicting human behavior during fires. Fire Technol 2013;49(1): 101–20.
[47] Yen HH, Lin CH, Tsao HW. Time-aware and temperature-aware fire evacuation path algorithm in IoT-enabled multi-story multi-exit buildings. Sensors 2021;21(1): 111.
[48] Eastman C, Teicholz P, Sacks R, Liston K. BIM handbook: a guide to building information modeling for owners, managers, designers, engineers, and contractors. John Wiley & Sons; 2018.
[49] Luo YX, Li Q, Jiang LR, Zhou YH. Analysis of Chinese fire statistics during the period 1997–2017. Fire Saf J 2021;125:103400.
[50] Siyam N, Alqaryouti O, Abdallah S. Research issues in agent-based simulation for pedestrians evacuation. IEEE Access 2020;8:134435–55.
[51] Deng Q, Zhang B, Zhou Z, Deng HY, Zhou L, Zhou Z, Jiang H. Evacuation time estimation model in large buildings based on individual characteristics and real-time congestion situation of evacuation exit. Fire-Switzerland 2022;5(6):204–23.
[52] Jiang CS, Deng YF, Hu C, Ding H, Chow WK. Crowding in platform staircases of a subway station in China during rush hours. Saf Sci 2009;47(7):931–8.
[53] Ding N, Chen T, Zhang H. Simulation of high-rise building evacuation considering fatigue factor based on cellular automata: a case study in China. Build Simul 2017; 10(3):407–18.
[54] Zhang N, Liang Y, Zhou CF, Niu MM, Wan F. Study on fire smoke distribution and safety evacuation of subway station based on BIM. Appl Sci-Basel 2022;12:19.
[55] Zhu R, Lin J, Becerik-Gerber B, Li N. Influence of architectural visual access on emergency wayfinding: a cross-cultural study in China, United Kingdom and United States. Fire Saf J 2020;113:102963.
[56] Nguyen MH, Ho TV, Zucker JD. Integration of smoke effect and blind evacuation strategy (SEBES) within fire evacuation simulation. Simul Model Pract Theory 2013;36:44–59.
[57] Ronchi E, Colonna P, Gwynne SMV, Purser DA. Representation of the impact of smoke on agent walking speeds in evacuation models. Fire Technol 2013;49(2): 411–31.
[58] Schröder B, Arnold L, Seyfried A. A map representation of the aset-rset concept. Fire Saf J 2020;115:103154.
[59] Zhang J, Zhu J, Dang P, Wu J, Zhou Y, Li W, Fu L, Guo Y, You J. An improved social force model (ISFM)-based crowd evacuation simulation method in virtual reality with a subway fire as a case study. Int J Digit Earth 2023;16(1):1186–204.
[60] Yang X, Zhang R, Li Y, Pan F. Passenger evacuation path planning in subway station under multiple fires based on multiobjective robust optimization. IEEE Trans Intell Transp Syst 2022;23(11):21915–31.
[61] Yang X, Zhang R, Pan F, Yang Y, Li Y, Yang X. Stochastic user equilibrium path planning for crowd evacuation at subway station based on social force model. Physical A 2022;594:127033.